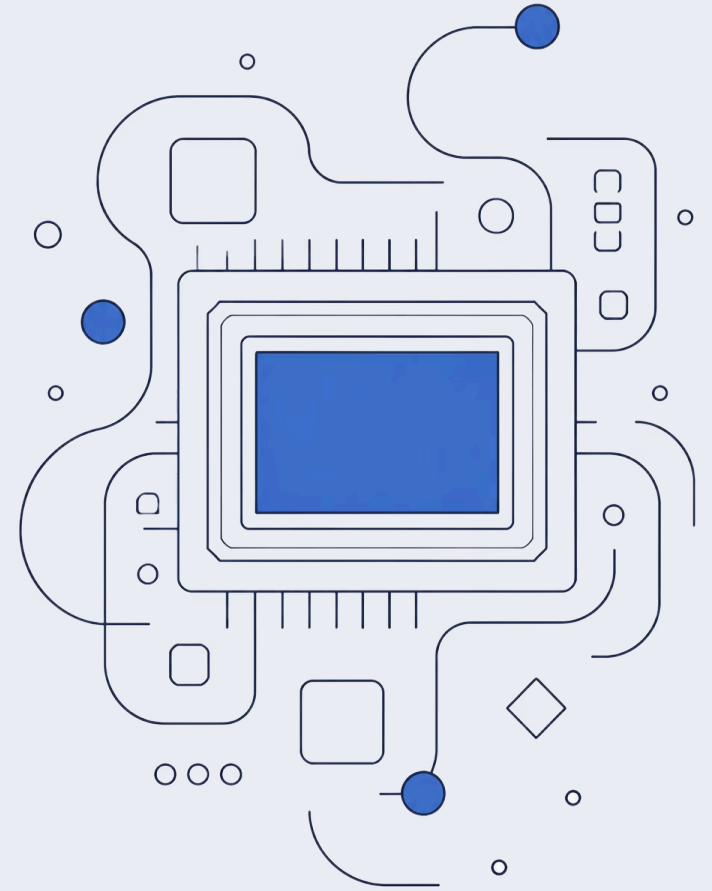


CCD vs. CMOS

The Evolution of Digital Imaging

From a Bell Labs experiment to the sensors in every smartphone — the story of how two technologies shaped the way we capture the world.



The Dawn of Digital Capture: The CCD Era

1

1969

Willard Boyle and George Smith invent the Charge-Coupled Device at Bell Labs — originally conceived as a data storage mechanism, not an imaging tool.

2

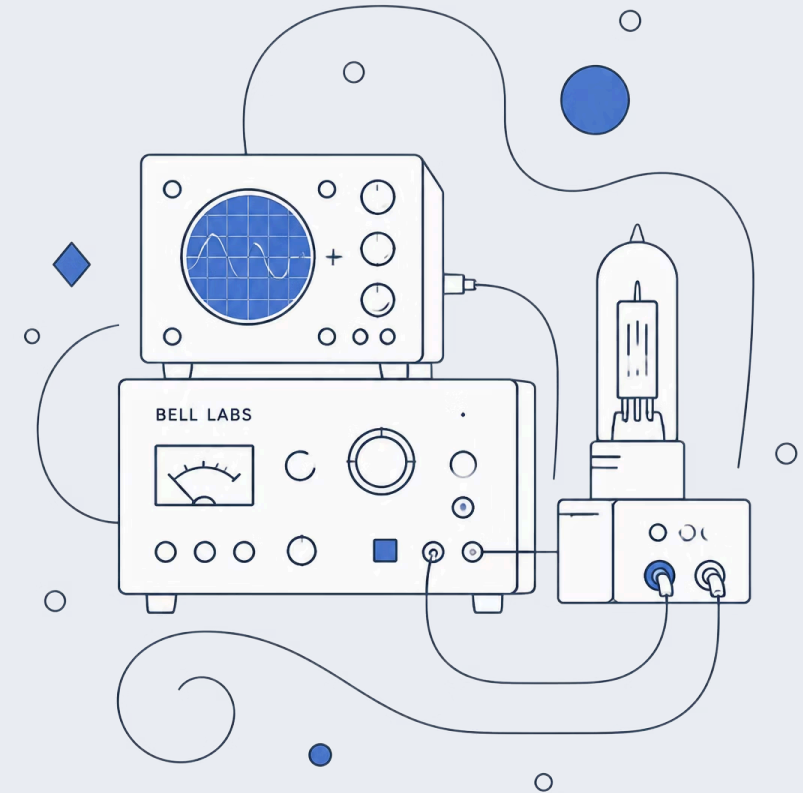
Early 1970s

Researchers quickly recognise the CCD's potential for light-sensitive imaging, transforming it from memory chip to camera sensor.

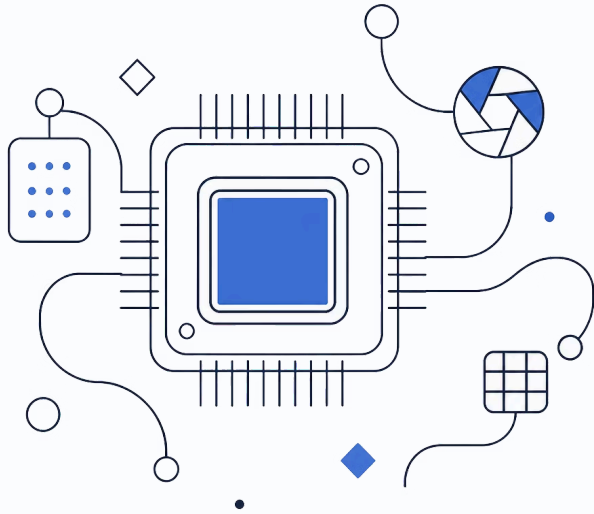
3

1976

The KH-11 KENNAN spy satellite launches, becoming the first spacecraft to use CCD array technology for real-time imaging.



CCD: The Gold Standard for Precision



How It Works

A CCD is an array of connected capacitors that shift electrical charge across the chip — like a **bucket brigade** — before converting it to a digital signal at a single output node.

Why It Dominated

High-quality, low-noise performance made CCD the benchmark for scientific, medical, and industrial imaging for decades.

- ❏ Kazuo Iwama of Sony championed CCD mass production so passionately that a CCD chip was immortalised on his tombstone in 1982.

The Rise of the Challenger: CMOS

Complementary Metal-Oxide-Semiconductor

Unlike CCD, CMOS sensors leverage **existing microprocessor fabrication processes** — the same technology used to build CPUs — making them cheaper and easier to scale.



Low Power

Each pixel has its own amplifier, consuming far less energy than CCD.

High-Speed Readout

Parallel readout enables faster frame rates.

On-Chip Integration

Signal processing can be built directly onto the sensor die.

The Conflict: Performance vs. Efficiency

CCD — The Precision Artist

- Superior light sensitivity and uniform signal quality
- Minimal noise, ideal for scientific imaging
- No blooming artefacts under bright light
- Dominant in high-end industrial and medical applications

CMOS — The Pragmatic Challenger

- Compact, low power, and significantly faster
- Early struggles with high noise and lower sensitivity
- Suffered from rolling shutter distortion
- Found early adoption in budget and consumer electronics



THE TURNING POINT

2015: The Industry Watershed Moment

In March 2015, Sony — the world's largest image sensor manufacturer — announced the discontinuation of its entire CCD sensor series.

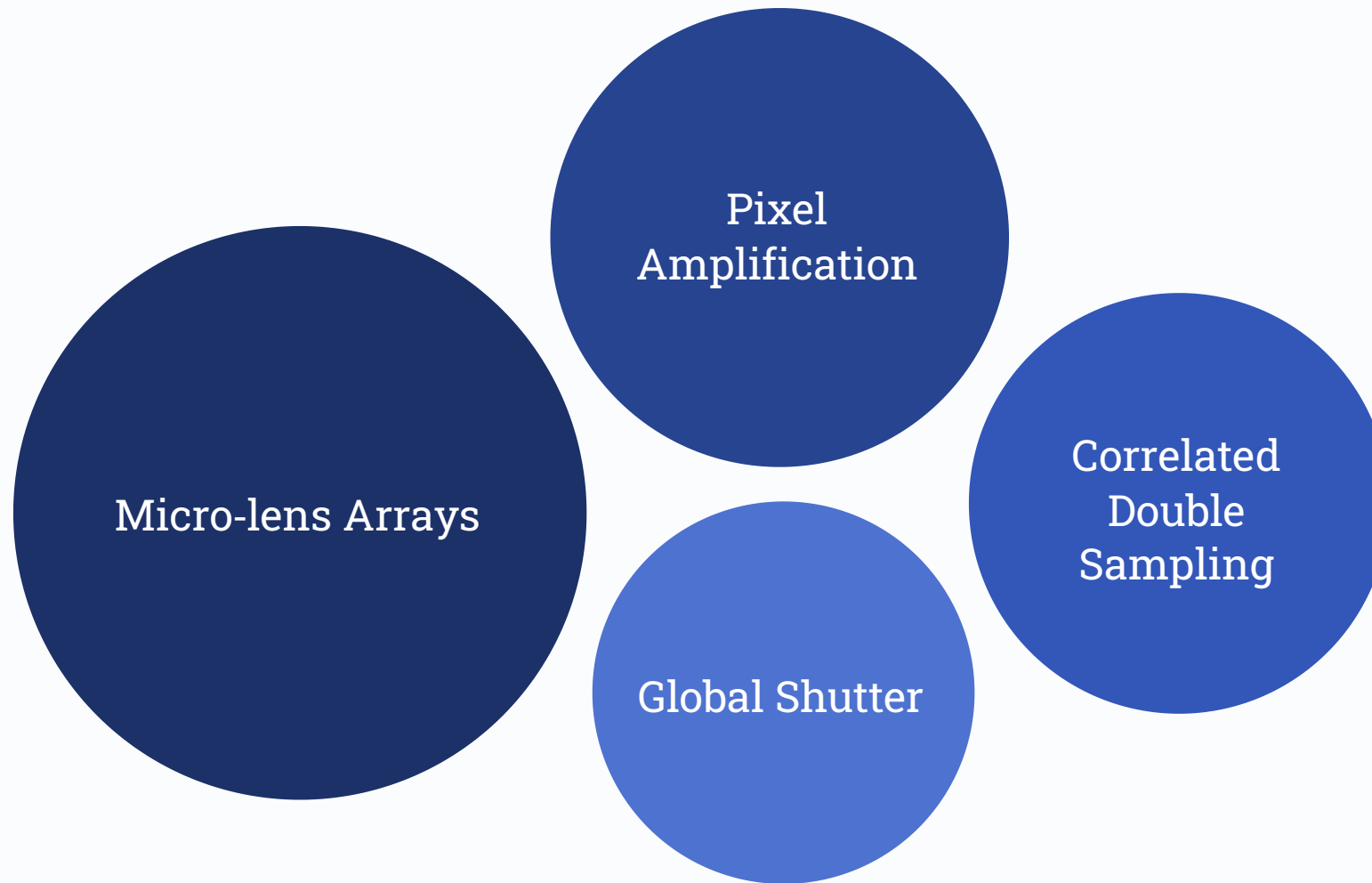
This decision marked the effective **end of new CCD development** worldwide. The "David vs. Goliath" battle concluded not with a single blow, but through CMOS's relentless scaling, cost reduction, and rapid performance gains. The industry had chosen its future.



Sony Headquarters Tokyo

ANNOUNCEMENT

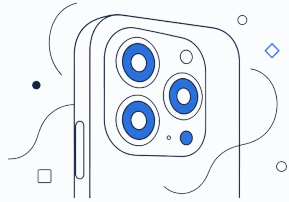
How CMOS Closed the Gap



These breakthroughs transformed CMOS from a budget compromise into a **technologically superior platform**, addressing every weakness that had previously favoured CCD.

TODAY

The Modern Era: CMOS Dominance



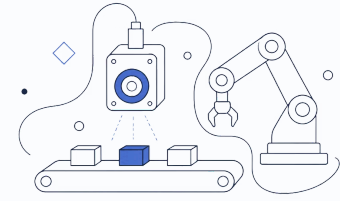
Smartphones

Billions of CMOS sensors power mobile photography, enabling computational imaging and AI-enhanced capture.



Autonomous Vehicles

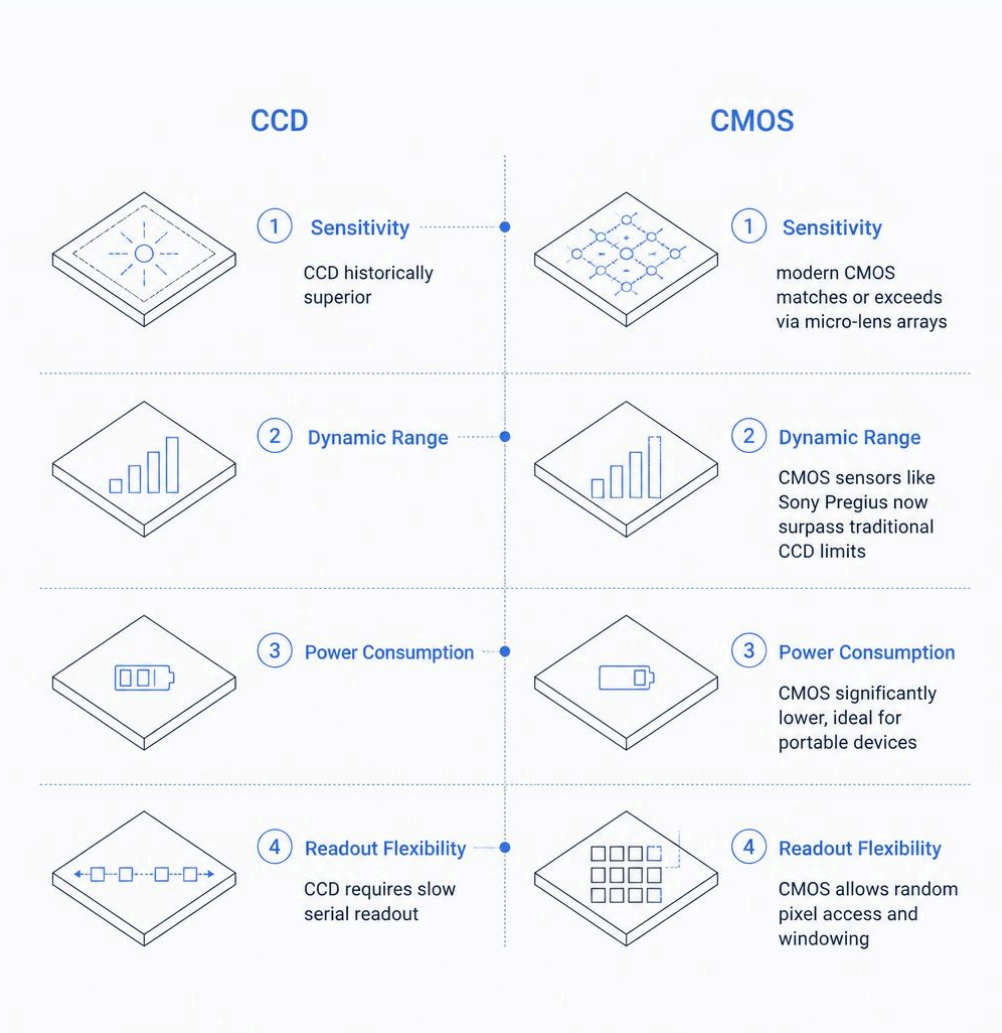
High-speed, low-latency CMOS sensors are critical for real-time environmental perception in self-driving cars.



Industrial Vision

System-on-a-chip integration enables smart cameras with on-board processing for high-speed inspection and robotics.

Key Technical Comparisons



The Numbers Tell the Story

Modern CMOS sensors have not merely caught up — in most measurable dimensions, they now **outperform** their CCD predecessors.

→ **Sensitivity:** Advanced fabrication and micro-lens arrays restore and exceed CCD light-gathering capability.

→ **Dynamic Range:** Sony Pregius and similar CMOS sensors surpass traditional CCD limits.

→ **Flexibility:** Random pixel access and region-of-interest readout impossible with CCD architecture.

CONCLUSION

The Legacy and the Future

CCD's Legacy

CCD provided the foundational technology for the digital imaging revolution — setting the quality benchmark that drove decades of innovation.

CMOS's Victory

CMOS delivered the scalability, integration, and efficiency that powers our hyper-connected visual world — from smartphones to space telescopes.

The Broader Lesson

Technological incumbents are eventually surpassed by versatile, integrated architectures that scale with the demands of their era.

